

CASE STUDY

Problem

Many businesses keep customer and lead contacts in Excel. Before they can send a WhatsApp offer or update, someone has to: - clean the phone numbers by hand (050 123 4567 , +971-50-... , 00971... , blank cells, two numbers in one cell) - remove the same person entered twice - work out who agreed to receive WhatsApp messages and who asked not to be contacted - copy and paste the same message with each person's name - keep some record of who received it, who didn't, and why

It's repetitive, easy to get wrong, and risky. Messaging people who opted out can get a WhatsApp business number restricted.

Solution

SheetReach turns that into one guided flow: **Contacts** → **Clean** → **Review** → **Message** → **Process** → **Results**. - **Upload** an Excel or CSV file. Columns are found automatically, even with title rows above the table. - **Clean**: every number is converted to international format, duplicates are removed on the *cleaned* number, and invalid or missing numbers are flagged. - **Review**: a business user sees the result in plain language before anything is saved. In the sample file: 1,250 rows → 1,092 ready, 41 duplicates, 15 invalid, 18 without a number, 12 on Do Not Contact and 72 that still need an opt-in. - **Message**: write once with `{{first_name}}` , `{{company}}` and `{{city}}` , with a live preview and safe fallbacks for missing data. - **Process**: a queue sends one message at a time. Temporary WhatsApp errors are retried and the campaign can be paused and resumed. - **Results**: sent, delivered, failed, skipped and opted out, each with a reason, plus an Excel export.

Automation (n8n)

A real, importable n8n workflow (`portfolio/n8n/whatsapp-excel-automation.json`) runs the whole process without anyone opening the app:

```
new contact file → upload & clean in SheetReach → data-quality gate → import clean contacts →  
Do Not Contact & opt-in check → pick message → create personalised campaign → start sending  
→ wait & check progress (loop) → download Excel results → final report
```

n8n calls SheetReach's API with standard HTTP Request nodes. The rules stay in one place (SheetReach), and n8n handles triggers, sequencing and reporting. The workflow ends cleanly with a reason if the file has nothing new or nobody is eligible. It was executed successfully in n8n 2.42.4 against the running app.

WhatsApp

SheetReach sends through a **provider layer**: - **Sandbox WhatsApp provider** (default, for testing): runs the full flow (acceptance, delivery updates, failures, retries and STOP replies) without contacting real numbers. Every campaign screen shows which provider is active. - **WhatsApp Business Cloud API provider** (production): built on Meta's official API, using approved message templates and the delivery-status webhook. It switches on only when the client's Meta credentials are configured. It was not run against Meta in this project.

No WhatsApp Web automation, browser bots, anti-ban tricks or rate-limit evasion are used.

Safety

- Only contacts with a WhatsApp opt-in on record can enter a campaign. A file without an opt-in column needs explicit confirmation.
- The **Do Not Contact** list is keyed by phone number. It is fed by STOP replies, WhatsApp's "user stopped marketing messages" error, "No" in the file's opt-in column and manual entries.
- The list is checked again **right before every message**, so someone who opts out mid-campaign is not messaged later in it, and is blocked from all future campaigns.
- Every skip and failure is recorded with a reason, and an activity log keeps the history.

Technical architecture (short)

Part	What it is
Web app	React + TypeScript + Tailwind, in the same design system as the rest of the portfolio series
API & rules	Python FastAPI: import/cleaning, eligibility, campaign queue, reporting
Database	SQLite by default; Postgres through one setting
WhatsApp	One provider layer: sandbox provider / official WhatsApp Cloud API provider
Automation	n8n workflow over the REST API
Quality	41 backend tests, a 33-step browser test of the full journey, and a real n8n execution

Result

What the workflow delivers, measured on a 1,250-row sample file with the sandbox provider: - A messy 1,250-row Excel file is cleaned and classified automatically, with a reason for every excluded row. - One message is personalised for 1,092 eligible contacts. The 72 without an opt-in are skipped and listed, not messaged. - The complete process runs end to end from

n8n, in about 1 minute, and returns a report with the Excel results. - Opt-outs during a campaign are added to Do Not Contact and excluded from the next campaign.

Delivery figures come from the sandbox provider; live delivery depends on the client's WhatsApp account and audience.

Screens

SheetReach
EXCEL → WHATSAPP

OVERVIEW

Dashboard

CONTACTS

Upload Excel

Contacts

Contact Collection

MESSAGING

Messages

Campaigns

SYSTEM

Activity Log

WhatsApp & n8n

WhatsApp connection
Sandbox provider - Cloud API ready

EXCEL → WHATSAPP AUTOMATION

Sandbox - WhatsApp Cloud API ready

STEP 1 · CONTACTS

Upload Excel

Upload your contact list. We clean it automatically and show you what's usable before anything is saved.

1 Upload file — 2 Review & clean — 3 Done

sample-contacts-1250.xlsx · 9 columns · 1,250 rows

ROWS IN FILE
1,250

READY TO MESSAGE
1,092

NEED OPT-IN
72

DUPLICATES REMOVED
41

INVALID NUMBERS
15

NO NUMBER
18

DO NOT CONTACT
12

ALREADY SAVED
0

1,164 new contacts will be saved (1,092 ready to message, 72 need opt-in). Problem rows are left out.

Cancel Import 1,164 contacts

Columns we found

Detected automatically. Change one only if it looks wrong.

FULL NAME
Contact Name
e.g. Hassan Fernandes

BUSINESS
Business Name
e.g. Al Noor Store

PHONE
Mobile
e.g. 054-604-4471

COUNTRY
Country
e.g. UAE

EMAIL

CITY

WhatsApp opt-in

Opt-in is read from the WhatsApp Opt-in column. Contacts marked "No" go to the Do Not Contact list. Blank ones are saved as Needs opt-in and won't be messaged.

Personal / SheetReach — Excel to WhatsApp campaign (n8n)

Editor Executions Evaluations

1 · Get the contact file

Run manually, or POST ("https://.../contacts.xlsx") to the webhook (e.g. from Google Drive, a form or a CRM export).
Edit Settings first — in Docker use http://host.docker.internal:8080.

Run manually
Settings
Download contact file

2 · Clean, validate, protect

SheetReach detects columns, normalises numbers, removes duplicates and checks opt-in + the Do Not Contact list. Only clean contacts are imported.

Upload & clean in SheetReach
Data quality report
New contacts to import?
Import clean contacts
DNC & opt-in check
Anyone ready to message?

3 · Personalise & send

One message, personalised per contact. Sent through SheetReach's WhatsApp provider: the sandbox provider for testing or the official WhatsApp Business Cloud API when the client's credentials are configured.

Load message templates
Pick message
Create personalised campaign
Start sending (WhatsApp provider)

4 · Track & report

Posts progress until the queue is done, downloads the Excel results and builds the final report.

Wait
Check progress
Campaign finished?
Download results (Excel)
Final report

Execute workflow from Run manually

Clear execution

SheetReach

EXCEL → WHATSAPP

OVERVIEW

DASHBOARD

CONTACTS

Upload Excel

Contacts

Contact Collection

MESSAGING

Messages

Campaigns

SYSTEM

Activity Log

WhatsApp & n8n

WhatsApp connection

Sandbox provider · Cloud API ready

EXCEL → WHATSAPP AUTOMATION

Sandbox · WhatsApp Cloud API ready

← Campaigns

Catalogue update — 7 Oct, 17:32

Message: New catalogue announcement · created 7 Oct, 17:32

Export results

Campaign complete

Finished 7 Oct, 17:33 · 47 retried automatically

Started by n8n · Complete

Delivered 1,018 · Awaiting delivery 23 · Failed 51 · Skipped 72

Sandbox · WhatsApp Cloud API ready

Sandbox run · connect the WhatsApp Cloud API to send live.

PROCESSED

1,164

of 1,164 · incl. 72 skipped

SENT

1,041

Accepted by WhatsApp

DELIVERED

1,018

98% of sent

FAILED

51

Reasons below

SKIPPED

72

Not messaged

OPTED OUT

32

Now on Do Not Contact

Why messages failed

Number is not on WhatsApp or cannot receive messages 30

WhatsApp limited marketing messages to this person (gave up after 2 retries) 15

Recipient stopped marketing messages from your business 6

Why contacts were skipped

No WhatsApp opt-in on record 72

All 1164

Sent 1041

Delivered 1018

Failed 51

Skipped 72

Opted out 32

Search contact or number

CONTACT

MESSAGE

STATUS

RETRIES

UPDATED

P07 · ARCHITECTURE

Excel → SheetReach → WhatsApp, orchestrated by n8n

Business rules live in SheetReach. n8n runs the process through SheetReach's API. The WhatsApp provider is swappable: a sandbox provider for testing, and the official Cloud API for live sending.

1 Excel / CSV contact file

Upload in the app, or a new file picked up by n8n (Drive, email, form, CRM export)

2 SheetReach · import & clean

Column detection · phone numbers → +E.164 · duplicates removed · invalid / missing flagged

3 Validation · opt-in · Do Not Contact

Only consented, valid, non-blocked contacts can enter a campaign — re-checked before every send

4 Personalised campaign queue

{{first_name}} · {{company}} · {{city}} per contact · retries with backoff · pause / resume

5 Status tracking

Sent · delivered · failed · STOP replies → Do Not Contact (Meta status webhook in production)

6 Results & report

Campaign dashboard with reasons · Excel export · n8n final report

AUTOMATION LAYER

n8n workflow

Trigger: manual run or webhook with a file URL

Download file → upload to SheetReach

Data-quality gate (stops if no phone column)

Import clean contacts

DNC & opt-in check — stops if nobody is eligible

Pick message → create personalised campaign

Start sending

Poll progress until complete

Download Excel results

Build final report

Calls the SheetReach REST API with real HTTP Request nodes. It doesn't duplicate any cleaning or sending logic.

BUILT & RUNNING

BUILT, NEEDS CLIENT CREDENTIALS

SHEETREACH · EXCEL → WHATSAPP AUTOMATION